

**3. Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.**

### **Aim**

To implement the **ID3 Decision Tree algorithm** and classify a new sample using an appropriate dataset.

### **Software Required**

- Python 3.x
- Pandas
- Scikit-learn
- Jupyter Notebook

### **Procedure**

1. Load the training dataset.
2. Separate input attributes and target class.
3. Convert categorical attributes into numerical values.
4. Create a Decision Tree classifier using entropy.
5. Train the model.
6. Provide a new test sample.
7. Predict its class.

### **Code**

```
import pandas as pd

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier

data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain', 'Rain',
               'Overcast', 'Sunny', 'Sunny', 'Rain', 'Sunny', 'Overcast',
               'Overcast', 'Rain'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool',
                   'Cool', 'Mild', 'Cool', 'Mild', 'Mild', 'Mild',
                   'Hot', 'Mild'],
```

```

'Humidity': ['High', 'High', 'High', 'High', 'Normal', 'Normal',
             'Normal', 'High', 'Normal', 'Normal', 'Normal', 'High',
             'Normal', 'High'],
'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak', 'Strong',
         'Strong', 'Weak', 'Weak', 'Weak', 'Strong', 'Strong',
         'Weak', 'Strong'],
'Play': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes',
         'No', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'No']
}

df = pd.DataFrame(data)
le = LabelEncoder()
for col in df.columns:
    df[col] = le.fit_transform(df[col])
X = df.drop('Play', axis=1)
y = df['Play']
model = DecisionTreeClassifier(criterion='entropy', random_state=0)
model.fit(X, y)
sample = [[2, 1, 1, 0]]
prediction = model.predict(sample)
print("Predicted Class:", prediction[0])

```

### Sample Output

```

print("Predicted Class:", prediction[0])
... Predicted Class: 0
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py
warnings.warn(

```

### Result

Thus, the **ID3 Decision Tree classifier** was successfully implemented and used to classify a new sample.